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5.2.3 Existence of the LU factorization

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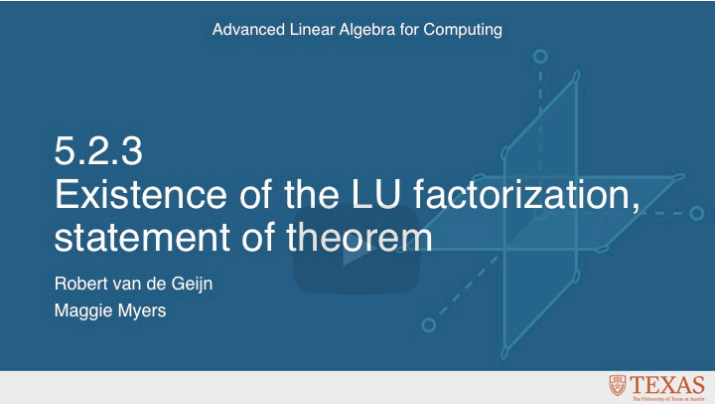
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We have derived an algorithm for LU factorization and we have linked that to the algorithm for Gaussian elimination, showing that they are equivalent. But something interesting comes up along the way, obviously. What

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Reading assignment

0.0/1.0 point (graded)



Read Unit 5.2.3 of the notes. [\[LINK\]](#)

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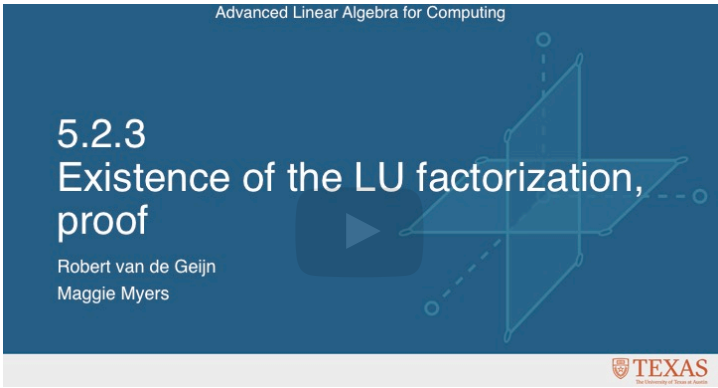
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equals $k + 1$. Okay, so let's start with the base case.



Now, if A only has one column, then it looks something like this alpha_11 over a_21. Okay, so this is now m by 1. Now it could be that a_21 is a vector of length 0. That's actually okay. Notice that this only has one principle leading submatrix. That's the matrix that's just alpha_11. So what we're saying is that that matrix is nonsingular. But if

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Homework 5.2.3.1

1/1 point (graded)
Prove Lemma 5.2.3.3.

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✓ Correct (1/1 point)

Homework 5.2.3.2

1/1 point (graded)
Choose the approximate cost incurred by

$$A = \text{LU-left-looking}(A)$$
$$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$$
$$A_{TL} \text{ is } 0 \times 0$$
$$\text{while } n(A_{TL}) < n(A)$$
$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$

Solve $L_{00}u_{01} = a_{01}$ overwriting a_{01} with u_{01}

$$\alpha_{11} := v_{11} = \alpha_{11} - a_{10}^T a_{01}$$
$$a_{21} := a_{21} - A_{20} a_{01}$$
$$a_{21} := l_{21} = a_{21} / \alpha_{11}$$
$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$
$$\text{endwhile}$$

when applied to an $m \times n$ matrix.

☒ $mn^2 - \frac{1}{3}n^3$

☐ $\frac{2}{3}n^3$

$\frac{1}{2}mn^2 - \frac{1}{6}n^3$



Solution:

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

Homework 5.2.3.4

1/1 point (graded)
Implement the algorithm

$$A = \text{LU-left-looking}(A)$$
$$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$$
$$A_{TL} \text{ is } 0 \times 0$$
$$\text{while } n(A_{TL}) < n(A)$$
$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$

$$\text{Solve } L_{00}u_{01} = a_{01} \text{ overwriting } a_{01} \text{ with } u_{01}$$
$$\alpha_{11} := v_{11} = \alpha_{11} - a_{10}^T a_{01}$$
$$a_{21} := a_{21} - A_{20} a_{01}$$
$$a_{21} := l_{21} = a_{21} / \alpha_{11}$$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$
$$\text{endwhile}$$

as

function [A_out] = LU_left_looking(A)

by completing the code in [Assignments/Week05/matlab/LU_left_looking.m](#). Input is an m \times n matrix A. Output is the matrix A that has been overwritten by the LU factorization. You may want to use [Assignments/Week05/matlab/test_LU_left_looking.m](#) to check your implementation.

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